

RetailIQ

E-Commerce Finance Analytics

End-to-End Project Report | Python • MySQL • Power BI

Project	RetailIQ — UK E-Commerce Finance Analytics
Dataset Period	January 2022 – December 2024 (orders); returns extend to January 2025
Tools	Python (pandas, NumPy, Matplotlib, Seaborn), MySQL 8.0, Power BI Desktop
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Prepared as a professional portfolio deliverable. All figures are sourced directly from the project files: the cleaned CSV datasets, the eight Python EDA notebooks, the six MySQL scripts (including embedded query outputs), and the published Power BI dashboard.

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1. Executive Summary

RetailIQ is an end-to-end retail analytics project built on a multi-table UK e-commerce dataset spanning January 2022 to December 2024 (with returns extending into January 2025). The work covers the full analytics lifecycle: profiling and cleaning a raw, deliberately messy eight-table dataset in Python, modelling it into a normalised MySQL schema with twelve foreign keys sharing a calendar dimension, diagnosing twelve business problems through SQL, and presenting the results in a seven-page Power BI executive dashboard.

The analysis identifies a coherent set of structural issues constraining profitability. Customer acquisition costs are rising sharply while lifetime value is falling, producing a collapse in the LTV-to-CAC ratio from 13.41 (2022) to 4.06 (2024). One-time buyers dominate the base at 68.8% of customers. Marketing budget is concentrated in the two least efficient channels — Paid Social and Google Ads absorb 84.3% of spend while delivering the weakest unit economics. Stockouts during peak season are estimated to have cost more than £1.5 million in lost revenue, with the 27in Monitor 4K alone accounting for £264,719. Returns are rising structurally (18.36% in 2022 to 21.42% in 2024) and are heavily concentrated in a single supplier, NovaTech Supply, whose 42.1% return rate is driven primarily by defective products. Month-end discounting is structural rather than promotional, costing £73,663 in 2022–2024 against only £8,081 for the rest of the month combined.

The report concludes with twelve business-problem deep dives and a set of evidence-based recommendations addressing each. The highest-leverage interventions identified are: reallocating marketing spend toward Email and Organic channels, enforcing product quality controls with NovaTech Supply (or restructuring that supplier relationship), raising inventory safety stock for premium high-velocity products, and replacing structural month-end discounting with targeted lifecycle promotions.

Headline KPI Snapshot

Metric	Value	Source
Total Revenue (completed orders)	£1,509,302.21	Dashboard card / kpis.sql
Total Orders (completed)	9,993	Dashboard / orders.csv
Average Order Value	£151.04	Dashboard card
Total Registered Customers	5,519	Dashboard card / customers.csv
Customers with Completed Orders	4,993	kpis.sql
Total Returns	3,716	Dashboard card / returns.csv
Stockout Lost Revenue (peak season)	£1.54M	Dashboard card / kpis.sql
Total Marketing Spend (2022–2024)	£877,294.83	Dashboard card / marketing_spend.csv
NovaTech Supply Return Rate	42.13%	Dashboard card / SQL Problem 10
Net Profit (line revenue – COGS – refunds)	£613,804.73	kpis.sql / verified against

		order_items.csv
Refund Amount (all returns)	£249,895.47	kpis.sql / returns.csv
Overall Return Rate	20.29%	kpis.sql
LTV-to-CAC Ratio (2024 cohort)	4.06	Dashboard card / SQL Problem 12

2. Project Overview

The RetailIQ project follows a layered analytics architecture, moving data from raw CSV files through a cleaning layer, a relational storage layer, an analytical SQL layer, and a presentation layer. Each layer is reproducible from the files in the repository.

Repository Structure

Folder / File	Contents	Role
dirty_dataset/	8 CSV files	Raw source data containing genuine data-quality issues
clean_dataset/	8 CSV files	Cleaned, analysis-ready output produced by the notebooks
eda_python/	8 Jupyter notebooks	One notebook per table — cleans and profiles each table
eda_sql/	6 SQL scripts	Schema creation, relationship and fix scripts, 18-problem EDA, executive KPIs
power_bi_dashboard/	dashboard.pbix, dashboard.pdf	Published Power BI report and PDF export

Analysis Pipeline

1. Each Python notebook loads its corresponding dirty CSV, profiles every column, applies targeted cleaning logic, and writes a cleaned CSV plus an audit file of flagged rows.
2. The eight cleaned CSVs are loaded into a MySQL 8.0 schema defined in `table_creation.sql`, with relationship and fix scripts (`relation.sql`, `fix_relation.sql`) adding twelve foreign keys, a marketing surrogate key, a derived date column, indexes and constraint repairs.
3. `main_sql_eda.sql` executes eighteen diagnostic SQL queries against the schema; `main_sql_eda_with_output.sql` documents the same queries with embedded result tables and written insights.
4. `kpis.sql` produces a single multi-CTE executive KPI query.
5. The Power BI report consumes the cleaned CSVs and presents the findings through KPI cards and visuals across seven pages.

3. Business Background

RetailIQ is a UK-based e-commerce retailer selling consumer electronics and accessories across twelve cities in England, Scotland and Wales. The product catalogue spans twelve categories — Accessories, Audio, Cameras, Displays, Furniture, Gaming, Lighting, PC Parts, Peripherals, Smart Home, Storage and Wearables — sourced from five suppliers: TechCore Ltd, NovaTech Supply, ValueSource Co, EliteComponents and BrightGadgets.

Customers are acquired through five marketing channels tracked in the marketing_spend table: Paid Social, Google Ads, Referral, Email and Organic. The cleaned customers table additionally contains a small number of customers labelled Direct and Unknown, which are not present in the marketing spend data. Orders pass through four statuses: Completed, Pending, Cancelled and Returned. The business operates a shared calendar dimension (date_dim) that supports consistent time-based analysis across all fact tables.

4. Problem Statement

Despite healthy top-line revenue of £1.51 million from completed orders over the three-year period, the underlying unit economics are deteriorating. The business faces a simultaneous squeeze on three fronts: rising acquisition costs, falling per-customer lifetime value, and growing return and refund pressure. Inventory shortages during peak demand periods are constraining revenue growth, while structural month-end discounting erodes margin without proportionally stimulating demand. Without intervention, continued spend growth will reduce profitability rather than increase it.

5. Business Objectives

This project was undertaken to:

- Cleanse a raw, multi-table retail dataset into analysis-ready tables suitable for relational modelling and reporting.
- Model a normalised MySQL schema that enforces referential integrity across eight tables sharing a calendar dimension.
- Diagnose twelve business problems covering churn, profit leakage, marketing ROI, stockouts, shipping economics, discounting, VIP customer management, returns, supplier quality, demand timing and unit economics.
- Produce a single-query executive KPI view and a Power BI dashboard that communicate the findings to a non-technical business audience.
- Translate the analysis into specific, evidence-based recommendations that directly address each business problem.

6. Dataset Overview

The project distinguishes between the dirty (raw) dataset and the cleaned dataset. The dirty dataset contains genuine data-quality issues; the cleaned dataset is the analysis-ready output produced by the Python notebooks and is the input to both the MySQL schema and the Power BI dashboard.

6.1 Dirty Dataset

The dirty dataset comprises eight CSV files organised as a star-style schema around a shared date_dim calendar table. Row counts below are taken directly from the raw CSV files in dirty_dataset/. Data-quality issues found in the raw data include:

- City typos — e.g. Londn, Londen, Manchestor, Liverpool, Leds, Leedes, Cardif, Glasgoe, Edinbugh.
- Supplier name variants in mixed case and spacing — e.g. novatech supply, Nova Tech Supply, NovaTech Supply; Techcore Ltd, TECHCORE LTD, Tech Core Ltd; Elite components, ELITECOMPONENTS; Bright gadgets, brightgadgets; Value Source Co, Valuesource Co.
- Type inconsistencies — age stored as float (31.0), cost_price as float (14.0), booleans encoded as the strings "True"/"False".
- Sentinel errors — qty = 999 and qty_returned = 999 used as placeholder error codes.
- Missing values across multiple columns including category, supplier, unit_price, cost_price, weight_kg, spend, impressions, clicks, cac, shipping_cost and return_reason.
- Negative values where only positive values are valid — negative refund_amount entries.
- Zero CAC rows in marketing_spend where spend and conversions were both non-zero.
- Inconsistent derived date fields (order_month, order_year, order_quarter, day_of_week) requiring recomputation and cross-checking against order_date.

Table	Rows (dirty)	Grain	Purpose
customers	5,522	1 row / customer	Demographics, acquisition channel, signup date, CAC
date_dim	1,096	1 row / day	Calendar attributes (year, quarter, month, weekend, month-end, peak season, season)
products	30	1 row / product	Product master — category, supplier, cost/sell price, margin, demand rank
orders	11,472	1 row / order	Order header — subtotal, shipping, total, discount, status
order_items	19,674	1 row / line item	Line economics — qty, unit price, discount, line total, cost price, weight
returns	3,791	1 row / return	Return reason, refund amount, qty returned
inventory	32,880	1 row / product / day	Daily stock level, units sold, stockout flag
marketing_spend	180	1 row / channel / month	Spend, impressions, clicks, conversions, CAC, budget share

6.2 Cleaned Dataset

The cleaned dataset retains the same eight-table structure but with all data-quality issues resolved. Row counts differ from the dirty dataset where rows were dropped as duplicates or removed during cleaning. All cleaned tables contain zero full-row duplicates.

Table	Rows (clean)	Rows (dirty)	Net Change	Notes
customers	5,519	5,522	-3	3 duplicate / dropped rows
date_dim	1,096	1,096	0	Boolean fields standardised; is_month_end recomputed
products	30	30	0	Suppliers unified to 5; cost_price/margin_pct repaired
orders	11,431	11,472	-41	Duplicates dropped; totals recalculated
order_items	19,642	19,674	-32	Duplicates dropped; line_total and discounted_price recalculated
returns	3,716	3,791	-75	Duplicates dropped; negative refunds corrected; 999 qty fixed
inventory	32,880	32,880	0	Date parsed; no row-level changes
marketing_spend	180	180	0	Missing spend/impressions/clicks imputed; zero CAC recomputed

Date Coverage

Table	Date Column	Earliest	Latest
date_dim	date	2022-01-01	2024-12-31
customers	signup_date	2022-01-01	2024-12-31
orders	order_date	2022-01-02	2024-12-31
order_items	order_date	2022-01-02	2024-12-31
inventory	date	2022-01-01	2024-12-31
returns	return_date	2022-01-14	2025-01-24

Returns extend into January 2025 because returned orders placed in late December 2024 are processed in the following weeks. The SQL return-rate analysis uses order_date (not return_date) so all 3,716 returns — including the 92 from January 2025 — are captured.

Order Status Distribution (Cleaned orders.csv)

Status	Order Count	Share of Orders
Completed	9,993	87.4%

Returned	649	5.7%
Cancelled	425	3.7%
Pending	364	3.2%

All SQL and Power BI revenue, profit and KPI analyses filter to orders with status = Completed unless explicitly stated otherwise. Product-profitability and return-rate analyses use order_items with order_status in (Completed, Returned) so that returned items remain in the denominator, producing an unbiased return rate.

7. Data Cleaning Process

Data cleaning was performed in Python using pandas, NumPy, Matplotlib and Seaborn. Each of the eight notebooks in `eda_python/` follows the same pattern: load the dirty CSV, inspect each column in turn, apply targeted cleaning logic, verify the fix, then write a cleaned CSV and (for orders) an issues CSV documenting rows that failed consistency checks. The cleaned outputs populate `clean_dataset/`.

7.1 Cleaning Summary by Table

Table	Notebook	Key Cleaning Actions
customers	eda_customers.ipynb	Trimmed and title-cased names; validated email format; fixed 16 city typos; filled missing city with mode and missing region via city→region mapping; set implausible ages (<0 or >120) to NaN; imputed missing age with gender-specific mean; standardised gender to M/F/Other and dropped rows with missing gender; parsed <code>signup_date</code> ; derived <code>signup_year</code> and <code>signup_month</code> ; audited CAC outliers by channel
date_dim	eda_date_dim.ipynb	Parsed date column; verified year, quarter, <code>month_num</code> , <code>month_name</code> , <code>week_num</code> , <code>day_of_week</code> , <code>is_weekend</code> , <code>is_month_end</code> and season all match the date; recomputed <code>is_month_end</code> using pandas' standard definition (last day of month only)
inventory	eda_inventory.ipynb	Stripped and validated <code>product_id</code> against <code>^P\d{3}\$</code> ; parsed date; verified no negative <code>stock_level</code> or <code>units_sold</code> ; confirmed <code>stockout_flag</code> is binary; computed per-product averages
marketing_spend	eda_marketing_spend.ipynb	Verified month format YYYY-MM; verified year and quarter consistency; imputed missing spend, impressions and clicks with channel medians; recomputed CAC where it was zero using <code>spend ÷ conversions</code> ; validated <code>budget_share</code>
orders	eda_orders.ipynb	Standardised column names; treated blank strings as NaN; parsed <code>order_date</code> and recomputed <code>order_month</code> , <code>order_year</code> , <code>day_of_week</code> , <code>order_quarter</code> ; mapped boolean strings to True/False; fixed 14 city typos; normalised <code>discount_pct</code> (>1 → /100); recalculated <code>shipping_cost</code> and <code>order_total</code> from components; added year/month/quarter/day-of-week/dedup consistency check flags and wrote <code>orders_issues.csv</code> ; dropped duplicates
order_items	eda_order_items.ipynb	Filled missing category and supplier via <code>product_id</code> mapping; standardised 24 supplier variants to 5 canonical names; replaced <code>qty=999</code> with the product's average qty and rounded to integer; filled missing <code>unit_price</code> with product median; recalculated <code>discounted_price</code> and <code>line_total</code> ; replaced zero <code>cost_price</code> with product median (or global median fallback); estimated <code>weight_kg</code> for 4 products with no weight data

products	eda_products.ipynb	Unified supplier names to 5 canonical values; stripped product_name and category; repaired P012 cost_price (set to 17.0) and recomputed its margin; reassigned P002 and P015 suppliers to NovaTech Supply; recomputed margin_pct for all products
returns	eda_returns.ipynb	Stripped and validated IDs; unified supplier variants to 5 canonical names; filled missing category and return_reason via product_id mapping; parsed return_date; converted negative refund_amount to absolute value; flagged suspiciously high refund-per-unit rows; replaced qty_returned=999 with 1

7.2 Supplier Standardisation

Across order_items, products and returns, the raw data contained 24 distinct supplier strings representing only 5 actual suppliers. The notebooks mapped every variant to its canonical form.

Canonical Supplier	Example Variants Found in Dirty Data
TechCore Ltd	Techcore Ltd, TECHCORE LTD, Tech Core Ltd, TechCore
NovaTech Supply	Novatech Supply, Nova Tech Supply, novatech supply, NOVATECH
ValueSource Co	Value Source Co, Valuesource Co, VALUE SOURCE, ValueSource
EliteComponents	Elite components, Elite Components, ELITECOMPONENTS
BrightGadgets	Bright gadgets, brightgadgets, BRIGHTGADGETS

7.3 Type Coercion and Missing-Value Treatment

Issue Type	Examples	Treatment
Float-to-int coercion	age 31.0 → 31; cost_price 14.0 → 14	Converted to int64 where appropriate
Boolean strings	is_weekend/is_month_end/is_peak_season as "True"/"False"	Mapped to Python bool / tinyint
Sentinel codes	qty=999, qty_returned=999	Replaced with product average (qty) or 1 (qty_returned)
Negative values	Negative refund_amount entries	Converted to absolute value
Zero CAC	marketing_spend rows where cac=0 but spend>0	Recomputed as spend ÷ conversions
Missing numerics	unit_price, cost_price, spend, impressions, clicks	Imputed with product / channel median; global median fallback
Missing categoricals	category, supplier, return_reason	Imputed via product_id mapping
Missing weights	weight_kg for P005, P011, P013, P021	Estimated from similar products (0.20, 0.80, 0.20, 0.70 kg)

Missing shipping_cost	Orders with NaN shipping_cost	Recalculated as order_total – (subtotal – discount_amount)
Implausible ages	age < 0 or age > 120	Set to NaN, then imputed with gender-specific mean

7.4 Consistency Auditing

The orders notebook recomputes derived date fields from order_date and flags mismatches using year_check, month_check, quarter_check and dow_check columns. It also flags shipping_cost_check, order_total_check and discount_check (discount_pct > 0.5). Rows failing any check are isolated into orders_issues.csv for review, and the final cleaned orders.csv is the audited, de-duplicated output. This audit-then-write approach means every analytical figure in the project can be traced back to a verifiable cleaning rule.

8. Exploratory Data Analysis Using Python

Eight standalone Jupyter notebooks perform per-table exploratory data analysis. Each notebook loads the dirty CSV, profiles every column (data type, null count, unique count, duplicates, descriptive statistics, value distributions, outliers via the IQR method), performs the cleaning actions described in Section 7, then re-profiles the cleaned data and exports it. The notebooks use pandas for manipulation, NumPy for numerical operations, and Matplotlib and Seaborn for visualisation.

8.1 Notebook Inventory

Notebook	Cells	Focus
eda_customers.ipynb	35	ID validation, demographics, city/region cleaning, channel and CAC profiling, CAC outlier audit by channel
eda_date_dim.ipynb	18	Calendar coverage, verification of every date-derived column against the date, seasonality and weekend flags
eda_inventory.ipynb	28	Stock-level and units-sold distributions, stockout_flag validation, per-product summaries
eda_marketing_spend.ipynb	25	Channel consistency, spend/impressions/clicks imputation, CAC recompute, budget-share validation
eda_orders.ipynb	31	City cleaning, discount normalisation, shipping/total recalculation, consistency auditing, duplicate removal
eda_order_items.ipynb	34	Category and supplier imputation, qty=999 fix, price recalculation, cost_price zero-replacement, weight estimation
eda_products.ipynb	24	Supplier unification, cost_price and margin_pct repair, demand-rank profiling
eda_returns.ipynb	34	Supplier standardisation, category and return_reason imputation, negative-refund correction, 999-qty fix, suspicious-refund flagging

8.2 Key Python EDA Findings

Table	Finding	Evidence
customers	5,519 customers after cleaning; age imputed to integer; 4 regions present (England, Scotland, Wales plus Unknown)	customers_cleaned.csv; cleaned age dtype int64
customers	16 distinct city typos corrected to 12 canonical UK cities (plus Unknown for unmappable rows)	city_typo_map in notebook cells 15–16
customers	CAC outliers identified per channel using IQR; Paid Social shows the highest absolute CAC values	IQR outlier block, cells 31–32
date_dim	All derived columns match the date after	Verification cells 6–13; fix in cell 14

	cleaning; is_month_end recomputed to standard pandas definition	
inventory	No negative stock or units; stockout_flag strictly binary; 32,880 rows retained	Notebook cells 17–22
marketing_spend	Missing spend/impressions/clicks imputed by channel median; zero-CAC rows recomputed	Cells 11–20
orders	14 city typos fixed; shipping_cost and order_total recalculated from components; duplicate rows dropped	Cells 10, 16, 19, 30
order_items	24 supplier variants unified to 5; discounted_price and line_total recalculated; 4 zero-cost products repaired	Cells 10, 24, 30
products	P012 cost_price corrected to 17.0 and margin recomputed; P002 and P015 reassigned to NovaTech Supply	Cells 11–14
returns	75 duplicate rows dropped; negative refunds converted to absolute; qty_returned=999 set to 1; suspicious refunds flagged	Cells 10, 29, 31

The Python EDA layer is the foundation of every subsequent analysis: the cleaned CSVs it produces feed the MySQL schema and the Power BI dashboard, so the integrity of every SQL finding and dashboard visual depends on the correctness of these notebooks.

9. Exploratory Data Analysis Using SQL

The SQL layer is implemented in MySQL 8.0 across six scripts in `eda_sql/`. The cleaned CSVs are loaded into a normalised schema, after which `main_sql_eda.sql` executes eighteen diagnostic queries. The companion `main_sql_eda_with_output.sql` documents the same queries with embedded result tables and written insights. `kpis.sql` produces a single executive KPI query using twelve common table expressions joined through CROSS JOIN.

9.1 Database Overview

The schema comprises eight tables with twelve foreign keys, all sharing the `date_dim` calendar dimension. Primary keys are defined inline in `table_creation.sql`; `relation.sql` adds the foreign keys, a surrogate auto-increment primary key for `marketing_spend` (which has no natural unique key) and a derived date column built from the month/year text columns; `fix_relation.sql` handles column type repairs, adds the `idx_order_date` index, repairs constraints and finalises the `dim_date` → `date_dim` rename.

Schema (Foreign Keys)

Child Table	Column	Parent Table	Column
orders	customer_id	customers	customer_id
orders	order_date	date_dim	date
order_items	order_id	orders	order_id
order_items	product_id	products	product_id
order_items	order_date	date_dim	date
returns	order_id	orders	order_id
returns	item_id	order_items	item_id
returns	product_id	products	product_id
returns	return_date	date_dim	date
inventory	product_id	products	product_id
inventory	date	date_dim	date
customers	signup_date	date_dim	date

`marketing_spend` additionally has a derived date foreign key to `date_dim.date`, added in `relation.sql` after the surrogate primary key and date column are created.

9.2 SQL Techniques Used

- CTEs (WITH clauses) for modular, readable multi-step queries — used in `kpis.sql` and most business-problem queries.
- Window functions — `NTILE(10)` for VIP decile segmentation (Problem 7), `NTILE(5)` for RFM scoring (Problem 13), `LAG` for month-over-month growth (Problem 15), and `SUM/AVG OVER (... ROWS BETWEEN 29 PRECEDING AND CURRENT ROW)` for rolling 30-day revenue (Problem 17).

- Self-joins — used for product affinity / co-purchase analysis (Problem 16).
- Multi-table joins across orders, order_items, returns, products, inventory, date_dim and marketing_spend.
- Conditional aggregation with CASE WHEN inside COUNT/SUM for cohort retention (Problem 14).
- TIMESTAMPDIFF for cohort month-number calculation.
- CROSS JOIN of twelve CTEs in kpis.sql to assemble a single executive KPI row.

9.3 SQL Analysis Summary (18 Problems)

#	Business Problem	Primary Tables
1	Silent customer churn — one-time buyers vs repeat	orders
2	Bestsellers generating negative true profit	order_items, returns
3	Marketing channels with worst ROI receiving most budget	marketing_spend, orders
4	Revenue lost to peak-season stockouts	inventory, date_dim, products
5	Cities/regions where shipping erodes profitability	orders
6	Discount addiction concentrated at month-end	orders
7	VIP (top 10%) customers and whether they get attention	orders, customers
8	Rising return rate over time and its financial impact	order_items, date_dim, returns
9	Low-value orders made unprofitable by shipping	orders
10	NovaTech Supply quality — excessive returns & refunds	order_items, returns
11	Weekend vs weekday demand gap	orders
12	CAC rising faster than LTV — collapsing unit economics	marketing_spend, customers, orders
13	RFM customer segmentation	orders, customers
14	Cohort retention analysis	orders
15	Month-over-month revenue growth	orders
16	Product affinity / cross-sell pairs	order_items
17	Rolling 30-day revenue trend	orders
18	Inventory turnover ratio & stock classification	inventory, products

Problems 1–12 are the headline business problems listed in the SQL script header and analysed in detail in Section 14. Problems 13–18 are supplementary analyses (RFM segmentation, cohort retention, MoM growth, product affinity, rolling revenue and inventory turnover) that extend the diagnostic coverage and feed several Power BI visuals.

9.4 Key SQL Findings

The following findings are taken directly from the embedded result tables and written insights in main_sql_eda_with_output.sql.

- Customer churn dominates the base: 68.8% of customers (3,437) buy once and never return, generating £531,971.53; the 7.1% with 6+ orders (355 customers) generate £472,886.01 — nearly matching the entire one-time segment.
- Two products run at a net loss after COGS and refunds: Noise Cancel Headphones (–£12,228.12) and Power Bank 20000mAh (–£6,929.30). Profit is driven by premium hardware: 27in Monitor 4K (+£101,562.18), Smart Watch Fitness (+£69,704.50), Mechanical Keyboard TKL (+£52,225.49).
- Marketing budget is misallocated: Email (LTV:CAC 33.41) and Organic (24.52) are the most efficient channels, but Paid Social (4.31) and Google Ads (4.65) absorb 84.3% of spend.
- Peak-season stockouts cost an estimated >£1.5 million; 27in Monitor 4K alone lost £264,719.18.
- Shipping erodes regional margin: Cardiff pays 17.1% of order value in shipping (worst); London delivers the highest volume (£538,096.27 from 3,580 orders) at only 8.9%.
- Discount addiction is structural: month-end average discount is 20.11% vs 0.72% elsewhere; month-end discount cost £73,663 vs £8,081 elsewhere — and it has grown every year (2022 → 2024).
- Unit economics are collapsing: LTV:CAC fell 13.41 (2022) → 9.05 (2023) → 4.06 (2024), as CAC rose £24.70 → £49.21 while LTV fell £331.26 → £199.65.
- NovaTech Supply is the quality outlier: 42.1% return rate vs 10–20% for other suppliers — 60.8% of its returns are "Defective/Not Working", a product-failure problem rather than a broad catalog issue.
- RFM shows value concentration: Champions (751 customers) hold £615,685.07 of revenue; 1,498 Lost customers still represent £233,686.73 of historical value; 500 At-Risk customers (LTV £481.17) are the prime reactivation target.
- No meaningful weekend demand uplift exists: weekend revenue is actually lower than the top weekday (Tuesday).
- Inventory turnover is high across all 30 products (all classified as Fast Mover), with the highest ratios for Gaming Mouse RGB (35.20), Power Bank 20000mAh (34.34) and Laptop Stand Aluminium (33.33); Accessories is the dominant category by turnover (349.97).

10. Power BI Dashboard Overview

The Power BI report (dashboard.pbix, exported as dashboard.pdf) presents the SQL findings as a seven-page executive dashboard. It consumes the eight cleaned CSVs directly and uses calculated DAX logic to reproduce the KPI definitions from kpis.sql. The dashboard is organised thematically: an executive overview, customer analytics, marketing analytics, returns analytics, geographic & shipping analytics, discount analytics, and product & inventory analytics.

10.1 Dashboard Pages and Visuals

Page	Visuals	Key Insights Shown
1 — Finance Overview	KPI cards (Total Revenue, Total Customers, AOV, Total Returns, Stockout Lost Revenue, Total Marketing Spend, NovaTech Return Rate, 2024 LTV:CAC); Cohort LTV vs Blended CAC by year; Quarterly Return Rate %	Headline executive view; LTV falling while CAC rising; return rate trending up
2 — Customer Analytics	Band Revenue by Order Count Band; Band Revenue by VIP Segment; Cohort LTV & Cohort Size by Signup Year; DOW Revenue; DOW AOV & Order Count; RFM segment table	68.8% one-time buyers; Non-VIP contributes 57.57% vs VIP 42.43%; Tuesday is the strongest revenue day
3 — Marketing Analytics	Channel LTV-to-CAC bar; Channel summary table (spend, budget share, CAC, LTV, ratio); CAC by Channel Year-by-Year; Customers Acquired vs Blended CAC by Year	Email/Organic most efficient; Paid Social/Google Ads dominate budget; CAC +99% over 3 years
4 — Returns Analytics	Supplier Return Rate %; Annual Refunds by return_reason (donut); NovaTech Product Returns; Annual Return Rate % & Annual Refunds Paid	NovaTech 42%; Defective/Not Working drives 44.72% of refunds; return rate 18.36% → 21.42%
5 — Geography & Shipping	City Revenue; Size Band Avg Shipping %; City Shipping Pct %; Size Band Orders	London dominates revenue; Cardiff worst shipping %; shipping burden regressive by basket size
6 — Discount Analytics	Period Discount Given; Yearly Discount Given by Year & Period; Discount Period summary table	Month-end discounting £73,663 vs £8,081 rest-of-month; growing year on year
7 — Product & Inventory	Product Net Profit; Inventory Turnover Ratio; Stockout Lost Revenue per product; Category Revenue; Category Stockout Rate %	Premium products drive profit; high-velocity products also have highest stockout rates

10.2 Dashboard-Reported KPIs and Figures

The following figures are read directly from the dashboard cards and visuals. Where a figure also appears in kpis.sql or the SQL output, the values reconcile.

KPI / Visual	Dashboard Value	Reconciles With
Total Revenue	£1.5093M	kpis.sql / orders.csv (Completed)
Total Customers	5,519	customers.csv (all registered)
Avg Order Value	£151.04	kpis.sql / orders.csv (Completed)
Total Returns	3,716	returns.csv
Stockout Lost Revenue	£1.54M	kpis.sql
Total Marketing Spend	£877.29K	marketing_spend.csv (2022–2024)
NovaTech Return Rate	42.13%	SQL Problem 10
LTV-to-CAC Ratio (2024 cohort)	4.06	SQL Problem 12 / kpis.sql
Cohort LTV per Customer (2022/2023/2024)	£331.26 / £315.64 / £199.65	SQL Problem 12
Blended CAC (2022/2023/2024)	£24.70 / £34.86 / £49.21	SQL Problem 12
Quarterly Return Rate range	15.22% – 22.46%	SQL Problem 8
Band Revenue (1 order / 2 / 3-5 / 6+)	£531,971.53 / £158,254.97 / £346,189.70 / £472,886.01	SQL Problem 1
VIP vs Non-VIP revenue share	42.43% / 57.57%	SQL Problem 7 (VIP £640,354; Non-VIP £868,947.89)
DOW Revenue (Tuesday top, Sunday lowest)	£232,427 / ... / £198,978	SQL Problem 11
Channel LTV-to-CAC (Email/Organic/Referral/Google/Paid Social)	33 / 25 / 11 / 5 / 4	SQL Problem 3
Annual Return Rate (2022/2023/2024)	18.36% / 19.91% / 21.42%	SQL Problem 8
Annual Refunds Paid (2022/2023/2024)	£45K / £83K / £122K	SQL Problem 8
NovaTech top return reasons (Defective / Not as Described)	44.72% / 16.21% of refunds	SQL Problem 10
NovaTech top returned products (Speaker / Earbuds / Plug / Watch)	524 / 505 / 301 / 158 returns	SQL Problem 10
City Revenue (London top, Nottingham)	£538,096.27 / ... / £23,833.80	SQL Problem 5

lowest)		
City Shipping % (Cardiff highest, Nottingham lowest)	17% / ... / 8%	SQL Problem 5
Size Band Avg Shipping % (Under £20 worst, £100+ best)	32.7 / 13.7 / 7.7 / 4.1	SQL Problem 9
Discount Given (Month-end vs Rest)	£73,663.32 vs £8,081.70	SQL Problem 6
Inventory Turnover Ratio (top: Gaming Mouse RGB)	35 / 34 / 33 / ...	SQL Problem 18
Stockout Lost Revenue per product (top: 27in Monitor 4K)	£0.26M / £0.19M / £0.12M / ...	SQL Problem 4

10.3 RFM Segment Figures (Power BI)

The Power BI RFM segment table on page 2 reports the following segment distribution (total 4,993 customers with completed orders). Note that the segment splits differ from the SQL output in `main_sql_eda_with_output.sql` because the dashboard's DAX measure applies a different NTILE banding; both views total 4,993 customers. The SQL segment figures are reported in Section 14.13.

Segment	Customers	Avg LTV	Total Revenue	Avg Orders	Avg Recency (days)
Champion	647	£888.29	£574,721.20	5.69	84.23
At Risk	503	£483.76	£243,329.90	3.17	707.77
Lost	1,500	£156.00	£233,996.60	1.00	753.17
Loyal	406	£392.31	£159,279.60	3.15	268.54
New Customer	1,214	£151.16	£183,507.20	1.00	102.92
Potential Loyal	723	£158.32	£114,467.80	1.00	372.94
Total	4,993	£302.28	£1,509,302.20	2.00	409.34

11. Key KPIs

The Key KPIs section combines the headline cards shown on dashboard page 1 with the additional executive KPIs produced by `kpis.sql`, all of which are verified against the cleaned datasets or dashboard visuals.

11.1 Dashboard Headline KPIs (Page 1 Cards)

KPI	Value	Definition / Source
Total Revenue	£1.5093M	Sum of <code>order_total</code> for Completed orders (<code>kpis.sql</code>)
Total Customers	5,519	All registered customers in <code>customers.csv</code> (dashboard card)
Avg Order Value	£151.04	Total revenue ÷ completed order count
Total Returns	3,716	Count of <code>return_id</code> in <code>returns.csv</code>
Stockout Lost Revenue	£1.54M	Estimated peak-season lost revenue from inventory + <code>date_dim</code> (<code>kpis.sql</code>)
Total Marketing Spend	£877.29K	Sum of spend across 2022–2024 (<code>marketing_spend.csv</code>)
NovaTech Return Rate	42.13%	NovaTech returns ÷ NovaTech items sold (SQL Problem 10)
LTV-to-CAC Ratio (2024)	4.06	2024 cohort LTV £199.65 ÷ blended CAC £49.21

11.2 Additional Executive KPIs (`kpis.sql`)

KPI	Value	Definition / Source
Total Customers (with completed orders)	4,993	Distinct <code>customer_id</code> with Completed orders (<code>kpis.sql</code>)
Total Orders (completed)	9,993	Count of Completed orders (<code>orders.csv</code>)
Net Profit	£613,804.73	<code>line_total</code> – (<code>cost_price</code> × <code>qty</code>) – refunds (<code>kpis.sql</code>)
Customer Lifetime Value (CLV)	£302.28	Avg revenue per customer with completed orders (<code>kpis.sql</code>)
Repeat Purchase Rate	28.19%	Share of customers with ≥ 2 completed orders (<code>kpis.sql</code>)
Return Rate	20.29%	Returns ÷ items fulfilled (Completed + Returned) (<code>kpis.sql</code>)
Refund Amount	£249,895.47	Sum of <code>refund_amount</code> across all returns (<code>kpis.sql</code>)
Average Shipping Cost %	—	Avg <code>shipping_cost</code> ÷ <code>order_total</code> for Completed orders (<code>kpis.sql</code>); see Section 14.5/14.9 for city- and

		band-level values
Month-End Discount Rate	20.11%	Avg discount_pct at month-end for Completed orders (kpis.sql)

Note on Total Customers: the dashboard card reports 5,519 (all registered customers), while kpis.sql defines Total Customers as the 4,993 distinct customers with at least one Completed order. Both figures are correct for their respective definitions and are reported here with clear labelling throughout.

12. Key Findings

Across the Python, SQL and Power BI layers, ten findings recur and form the evidence base for the recommendations in Section 16.

1. Churn dominates the customer base

68.8% of customers (3,437) buy once and never return, generating £531,971.53. The 7.1% with 6+ orders (355 customers) generate £472,886.01 — nearly matching the entire one-time segment. Average LTV scales steeply with frequency: £154.78 (1 order) → £284.12 (2) → £537.56 (3–5) → £1,332.07 (6+).

2. Two products run at a net loss

Noise Cancel Headphones (–£12,228.12) and Power Bank 20000mAh (–£6,929.30) generate negative net profit after COGS and refunds. Profit is driven by premium hardware: 27in Monitor 4K (+£101,562.18), Smart Watch Fitness (+£69,704.50), Mechanical Keyboard TKL (+£52,225.49).

3. Marketing budget is misallocated

Email (LTV:CAC 33.41) and Organic (24.52) are the most efficient channels but receive only 5.4% and 3.4% of spend respectively. Paid Social (4.31) and Google Ads (4.65) absorb 84.3% of spend despite delivering the weakest unit economics.

4. Stockouts are a bigger lever than acquisition

Peak-season stockouts cost an estimated >£1.5M in lost revenue. The 27in Monitor 4K alone lost £264,719.18; Smart Watch Fitness lost £192,826.87 and Mechanical Keyboard TKL £115,872.11.

5. Shipping erodes regional margin

Cardiff pays 17.1% of order value in shipping (worst); London delivers the highest volume (£538,096.27 from 3,580 orders) at only 8.9%. Scotland (Edinburgh 14.3%, Glasgow 13.5%) is consistently more expensive to serve than England.

6. Discount addiction is structural

Month-end average discount is 20.11% vs 0.72% the rest of the month. Month-end discount cost £73,663.32 against £8,081.70 elsewhere, and it has grown every year: £14,605 (2022) → £25,760 (2023) → £33,298 (2024).

7. Unit economics are collapsing

LTV:CAC fell 13.41 (2022) → 9.05 (2023) → 4.06 (2024), as blended CAC rose £24.70 → £49.21 while average cohort LTV fell £331.26 → £199.65. The 2024 cohort is near the warning threshold (<5×).

8. NovaTech Supply is the quality outlier

NovaTech's 42.1% return rate is more than double the next worst supplier. 60.8% of its returns are "Defective/Not Working". Two SKUs (Bluetooth Speaker Mini: 524 returns; Wireless Earbuds Pro: 505) drive the majority of its return volume; together they account for £64,566.50 in refunds.

9. Small orders are implicitly subsidised

Orders under £20 carry shipping at 32.7% of order value on average; 42 completed orders have shipping costs exceeding the product subtotal itself, and 491 orders have shipping above 50% of subtotal. Once subtotal exceeds £50, shipping drops below 10% of order value.

10. No meaningful weekend demand uplift

Weekend revenue (Sat + Sun $\approx$ £423,883) is lower than Tuesday alone (£232,427). Order counts are unusually flat across the week (~1,370–1,499 per day), indicating stable baseline demand rather than weekend-driven spikes.

13. Business Insights

The ten findings above combine into four cross-cutting insights that frame the recommendations.

Insight 1 — A three-layer margin squeeze

Profitability is being compressed simultaneously by rising acquisition costs (CAC +99% over three years), falling per-customer value (LTV -40% between 2022 and 2024 cohorts), and rising post-sale losses (return rate 18.36% → 21.42%, refunds nearly tripling from £44.7K to £122.0K). No single lever can resolve this; the response must be coordinated across marketing, product and operations.

Insight 2 — Growth is acquisition-led, not retention-led

Cohort sizes are growing (1,475 → 2,205 customers per year) but per-customer value is shrinking fast enough to offset the scale. Revenue growth is episodic and campaign-driven (MoM swings of ±20–40%) rather than compounding, and rolling 30-day analysis would show whether spikes are sustained or marketing-dependent. The business is structurally dependent on continued ad spend to maintain revenue.

Insight 3 — Value is concentrated but defensible

A small group of high-frequency customers (6+ orders, 355 customers; Champions, 647–751 depending on view) holds a disproportionate share of revenue. Retention decay happens quickly after Month 1–3 in cohort analysis, so the biggest leverage point is extending early-lifecycle engagement rather than acquiring more one-time buyers.

Insight 4 — Fixable supply-side issues distort the numbers

NovaTech Supply alone accounts for £101,210.97 of refunds — more than every other supplier combined in some quarters — and its return problem is overwhelmingly product-quality-driven (60.8% Defective/Not Working). Similarly, stockouts are concentrated in a small number of high-velocity premium products. Fixing these two supply-side issues would mechanically improve return rate, refund exposure and lost-revenue figures without any change to customer behaviour.

14. Business Questions Analysis

The SQL script header lists twelve business problems, each analysed below with the question, the analysis performed, key findings, the business insight, and a specific recommendation supported by the analysis. Each finding references the embedded result tables in `main_sql_eda_with_output.sql`.

14.1 Business Question 1

Question: Why are so many customers purchasing only once and never returning?

Analysis Performed

Grouped all customers with at least one Completed order by order-frequency band (1 order / 2 / 3–5 / 6+) and computed the customer count, percentage of base, average lifetime value and total revenue per band.

Key Findings

Order Count Band	Customers	% of Base	Avg LTV	Total Revenue
1 order (one-time)	3,437	68.8%	£154.78	£531,971.53
2 orders	557	11.2%	£284.12	£158,254.97
3–5 orders	644	12.9%	£537.56	£346,189.70
6+ orders	355	7.1%	£1,332.07	£472,886.01

Business Insight

One-time buyers dominate the base (68.8%) but contribute relatively little revenue, while the 7.1% of customers with 6+ orders generate nearly the same revenue as the entire one-time segment. Retention has a steep, non-linear impact on customer value.

Business Recommendation

Shift marketing weight from acquisition volume toward post-purchase retention: deploy a structured first-to-second-order programme (post-purchase email series, personalised 30-day repurchase offer, product-setup content) targeted at the 3,437 one-time buyers. The data shows that moving even a small fraction into a second order more than doubles their average LTV (£154.78 → £284.12), making the second-purchase moment the highest-ROI intervention point.

14.2 Business Question 2

Question: Which bestselling products are generating negative true profit after costs and refunds?

Analysis Performed

Joined `order_items` to `returns` on `item_id`, computed gross revenue, COGS (`cost_price × qty`), gross profit, total refunds and net profit per product, for items with `order_status` in (Completed, Returned). Sorted by net profit ascending.

Key Findings

Product	Category	Supplier	Net Profit
Noise Cancel Headphones	Audio	TechCore Ltd	−£12,228.12
Power Bank 20000mAh	Accessories	TechCore Ltd	−£6,929.30
Thermal Paste Premium	PC Parts	EliteComponents	£1,331.00
Cable Clips Pack 50	Accessories	EliteComponents	£1,668.23
— (highest profit) —			
Mechanical Keyboard TKL	Peripherals	EliteComponents	£52,225.49
Smart Watch Fitness	Wearables	NovaTech Supply	£69,704.50
27in Monitor 4K	Displays	TechCore Ltd	£101,562.18

Business Insight

Two products run at a net loss because COGS alone exceeds revenue before refunds are considered — indicating a pricing, sourcing or inventory-cost issue rather than a returns problem. Premium hardware (Displays, Wearables, Peripherals) drives the strongest profit contributions.

Business Recommendation

Immediately review the unit economics of Noise Cancel Headphones and Power Bank 20000mAh: renegotiate cost price with TechCore Ltd, raise selling price, or discontinue. For Smart Plug 4-Pack (gross profit £16,738.92 reduced to £4,499.53 net by £12,239.39 of refunds), prioritise defect reduction. Do not use revenue ranking alone for product decisions — profitability analysis should drive ranging and promotion choices.

14.3 Business Question 3

Question: Which marketing channel is receiving significant budget despite delivering the worst ROI?

Analysis Performed

Computed total spend, budget share, average CAC, customers acquired and average LTV per channel by joining marketing_spend (spend, cac, conversions) to completed orders (revenue per customer). Derived the LTV-to-CAC ratio per channel and sorted ascending.

Key Findings

Channel	Spend	Budget Share	Avg CAC	Customers Acquired	Avg LTV	LTV-to-CAC
Paid Social	£472,746.36	53.9%	£70.50	6,982	£304.19	4.31
Google Ads	£266,403.18	30.4%	£63.79	4,384	£296.57	4.65
Referral	£60,236.05	6.9%	£23.50	2,678	£249.40	10.61
Organic	£30,213.06	3.4%	£13.43	2,355	£329.36	24.52

Email	£47,696.18	5.4%	£10.07	4,954	£336.47	33.41
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Business Insight

Paid Social and Google Ads absorb 84.3% of the budget yet deliver the weakest LTV-to-CAC ratios (4.31 and 4.65). Email and Organic generate the highest customer LTVs (£336.47 and £329.36) at the lowest CACs, making them disproportionately profitable.

Business Recommendation

Reallocate marketing spend from Paid Social and Google Ads toward Email and Organic. A phased shift — for example moving 10–15 percentage points of budget share from Paid Social into Email lifecycle programmes and Organic SEO/content — would improve blended marketing ROI without reducing total spend. Set channel-level LTV:CAC floor targets (e.g. minimum 8×) and reduce spend on any channel that falls below the floor for two consecutive quarters.

14.4 Business Question 4

Question: How much revenue is being lost due to stockouts during peak seasonal demand?

Analysis Performed

Computed average daily units sold per product from inventory rows where `stockout_flag = 0`, counted peak-season stockout days per product by joining inventory to `date_dim` on `is_peak_season = 1`, then estimated lost revenue as `stockout_days × avg_daily_units × sell_price`.

Key Findings

Product	Stockout Days (peak)	Avg Daily Units	Est. Lost Revenue
27in Monitor 4K	—	6.22	£264,719.18
Smart Watch Fitness	138	9.86	£192,826.87
Mechanical Keyboard TKL	—	—	£115,872.11
Docking Station USB-C	—	—	£91,040.83
Bluetooth Speaker Mini	—	—	£77,884.18
— cumulative all products —			> £1.5 million

Business Insight

High-value premium products account for a disproportionate share of lost revenue: the top three alone represent more than £573K in estimated missed sales. Several modest-demand products still create major losses because of their price points. Inventory availability — not acquisition — appears to be the larger revenue lever.

Business Recommendation

Raise safety stock levels for the top five revenue-impact products (27in Monitor 4K, Smart Watch Fitness, Mechanical Keyboard TKL, Docking Station USB-C, Bluetooth Speaker Mini) ahead of each peak season.

Incorporate revenue impact, not just unit-velocity forecasts, into replenishment planning. Target a reduction of peak-season stockout days by 50% for these SKUs — recovering even a quarter of the £1.5M estimated loss would exceed the annual marketing budget reallocation in Q3.

14.5 Business Question 5

Question: Which cities or regions have shipping costs that are eroding profitability?

Analysis Performed

Grouped Completed orders by city and region, computing total orders, total revenue, average order value, average and total shipping cost, and shipping as a percentage of order value. Sorted by average shipping cost descending.

Key Findings

City	Region	Orders	Revenue	Avg Shipping	Shipping % of Order
Cardiff	Wales	277	£37,299.17	£11.01	17.1%
Edinburgh	Scotland	372	£56,634.21	£11.78	14.3%
Glasgow	Scotland	643	£103,928.40	£11.31	13.5%
Newcastle	England	376	£57,183.65	£8.60	11.5%
Manchester	England	1,138	£173,086.16	£7.49	9.3%
London	England	3,580	£538,096.27	£7.14	8.9%
Nottingham	England	145	£23,833.80	£8.52	8.4%

Business Insight

London is the dominant market (revenue greater than Manchester, Birmingham and Glasgow combined) and the most efficient to serve despite the highest volume, indicating scale economies. Cardiff (17.1%) and Scotland (Edinburgh 14.3%, Glasgow 13.5%) are the least efficient regions; geographic profitability differences are driven by fulfilment economics, not customer spending behaviour.

Business Recommendation

Introduce region-specific shipping policies: a minimum free-shipping threshold for Cardiff and Scottish postcodes (where shipping exceeds 13% of order value), and tiered shipping fees for low-basket orders in those regions. Investigate carrier contracts or regional fulfilment options for Scotland and Wales. Continue investing in London's high-volume, low-cost delivery network, but monitor revenue-concentration risk.

14.6 Business Question 6

Question: Is the business becoming overly dependent on discounts, especially at month-end?

Analysis Performed

Split Completed orders into month-end (days 24–31) vs rest of month, computing order count, average discount percentage, average order value, total revenue and total discount given. Repeated the split by order_year to reveal the trend.

Key Findings

Period	Orders	Avg Discount %	Avg Order Value	Total Revenue	Total Discount Given
Month-end (24–31)	2,445	20.11%	£127.21	£311,040.01	£73,663.32
Rest of month (1–23)	7,548	0.72%	£158.75	£1,198,262.20	£8,081.70

Business Insight

Month-end discounting is structural, not promotional: the ~20% rate is tightly clustered across all three years (19.95% / 20.24% / 20.08%) while the financial cost scales rapidly (£14.6K → £25.8K → £33.3K) as month-end order volume rises. Discounts are not stimulating proportional revenue — month-end AOV is actually lower than rest-of-month.

Business Recommendation

Replace blanket month-end discounting with targeted lifecycle promotions: rather than a flat ~20% at month-end, offer smaller, behaviour-triggered discounts (e.g. basket-abandonment recovery, post-purchase cross-sell, VIP-only previews). Set a monthly discount-cost cap and track discount revenue-efficiency (discount £ spent per incremental revenue £). The current pattern indicates discounts are being used to clear inventory or hit targets — redirect that pressure toward demand smoothing and inventory planning.

14.7 Business Question 7

Question: Who are the highest-value (VIP) customers, and are they receiving sufficient attention?

Analysis Performed

Ranked all customers with Completed orders by lifetime value using NTILE(10) into deciles; decile 1 = VIP (top 10%). Produced both the full VIP customer list (with first/last order date and days since last order) and a VIP vs Non-VIP summary.

Key Findings

Segment	Customers	Avg Orders	Avg LTV	Total Revenue
VIP (top 10%)	500	6.8	£1,280.71	£640,354.32
Non-VIP	4,493	1.5	£193.40	£868,947.89

Business Insight

VIPs generate 6.6× higher lifetime value and 4.5× more orders on average than Non-VIPs, but revenue concentration is moderate rather than extreme (VIPs 42.43% vs Non-VIPs 57.57% of revenue per the dashboard). The Non-VIP segment is the real volume engine — the biggest upside is migrating Non-VIPs upward, not just protecting VIPs.

Business Recommendation

Establish a formal VIP programme with dedicated benefits (priority support, early access, exclusive previews, dedicated account contact for the top 500), and simultaneously build a mid-tier migration programme to convert high-frequency Non-VIPs into VIPs. Monitor the days-since-last-order metric for VIPs — outliers beyond ~250 days are churn-risk candidates within the high-value segment and should trigger a personalised reactivation contact.

14.8 Business Question 8

Question: Is the product return rate increasing over time, and what is its financial impact?

Analysis Performed

Joined order_items to date_dim on order_date and to returns on item_id, computing items fulfilled, items returned, return rate percentage and refunds paid per quarter and per year. Used order_date (not return_date) so all 3,716 returns — including the 92 from January 2025 — are captured.

Key Findings

Year	Items Fulfilled	Items Returned	Return Rate %	Refunds Paid
2022	3,665	673	18.36%	£44,665.12
2023	6,298	1,254	19.91%	£83,228.92
2024	8,351	1,789	21.42%	£122,001.43

Business Insight

Return rates are worsening structurally (18.36% → 19.91% → 21.42%) and refund costs are compounding faster than return rates (nearly tripling while the rate rises only ~3 points). Q4 is consistently the worst quarter each year, and 2024 stays above 20% every quarter — indicating deterioration, not fluctuation.

Business Recommendation

Treat rising returns as a margin-equivalent issue to CAC inflation. Combine the supplier-quality fix in Q10 with category- and channel-level return-rate dashboards to identify whether returns are driven by product quality, marketing-quality mismatch or fulfilment errors. Set a return-rate ceiling target (e.g. ≤19% within 12 months) and tie supplier scorecards to it. Expect returns to compound with discount and stockout pressure if left unaddressed.

14.9 Business Question 9

Question: Which low-value orders have shipping costs that make them unprofitable?

Analysis Performed

Grouped Completed orders by subtotal band (Under £20, £20–49, £50–99, £100+) and computed order count, average subtotal, average shipping cost, average shipping as % of total, count of orders where shipping ≥ subtotal, and count where shipping ≥ 50% of subtotal. Also listed the 20 worst individual orders.

Key Findings

Order Band	Orders	Avg Subtotal	Avg Shipping	Shipping %	Ship ≥ Subtotal	Ship ≥ 50% Subtotal
A: Under £20	1,136	£11.30	£5.17	32.7%	42	491
B: £20–49	2,031	£37.57	£5.75	13.7%	0	1
C: £50–99	2,242	£75.75	£6.10	7.7%	0	0
D: £100+	4,584	£255.07	£10.78	4.1%	0	0

Business Insight

Shipping burden is heavily regressive: orders under £20 carry shipping at 32.7% of order value, and 42 orders have shipping exceeding the entire product subtotal. The worst individual orders are single-item purchases in Cardiff, Glasgow, Edinburgh and Newcastle. The core business is healthy at scale — profitability issues are concentrated at the bottom of the basket-size distribution.

Business Recommendation

Introduce a minimum order value for free shipping (e.g. £30) and tiered shipping fees for baskets below that threshold. Add bundling incentives ("frequently bought together" modules — supported by the Q16 affinity analysis) to lift single-item baskets above the unprofitable shipping zone. Track the count of orders where shipping ≥ subtotal as a weekly operational KPI; the current 42 should reduce to zero under the new policy.

14.10 Business Question 10

Question: Is NovaTech Supply causing excessive returns, refunds, and quality-related issues?

Analysis Performed

Computed return rate and total refunds per supplier across all order_items with order_status in (Completed, Returned); then drilled into NovaTech's return reasons and the specific NovaTech products driving returns.

Key Findings

View	Result
NovaTech return rate	42.1% (3,532 items sold, 1,488 returned)
Next-worst supplier (ValueSource)	20.1%
Healthy range (TechCore, Elite)	13.1% – 13.4%

NovaTech total refunds	£101,210.97
Top return reason (Defective)	905 returns / 60.8% / £60,563.90 refunds
Second (Not as Described)	192 returns / 12.9% / £13,070.00 refunds
Top returned SKU (Bluetooth Spk)	524 returns / £35,484.69 refunds
Second SKU (Wireless Earbuds Pro)	505 returns / £29,081.81 refunds

Business Insight

NovaTech is a categorical, not marginal, outlier: a 42.1% return rate more than double the next supplier, with 60.8% of returns due to "Defective/Not Working" — a product-failure problem, not a logistics or expectation-mismatch problem. Two SKUs (Bluetooth Speaker Mini and Wireless Earbuds Pro) drive the majority of the return volume.

Business Recommendation

Issue a formal supplier quality improvement plan to NovaTech Supply with quantified defect-rate targets and a 90-day review. Pending improvement, reduce or suspend orders of the two highest-return SKUs (Bluetooth Speaker Mini and Wireless Earbuds Pro) and source alternatives from TechCore Ltd or EliteComponents, whose return rates sit in the 13–14% healthy range. Fixing NovaTech alone would materially reduce the overall return rate more than broad operational improvements across all other suppliers.

14.11 Business Question 11

Question: Is there a meaningful difference in demand between weekends and weekdays?

Analysis Performed

Grouped Completed orders by day_of_week and is_weekend, computing order count, average order value and total revenue per day.

Key Findings

Day	Weekend	Orders	Avg Order Value	Total Revenue
Saturday	Yes	1,401	£160.53	£224,904.63
Sunday	Yes	1,400	£142.13	£198,978.47
Tuesday	No	1,499	£155.05	£232,426.99
Thursday	No	1,468	£148.29	£217,683.70
Friday	No	1,438	£150.34	£216,188.52
Monday	No	1,417	£150.13	£212,736.99
Wednesday	No	1,370	£150.64	£206,382.91

Business Insight

There is no meaningful weekend uplift: combined weekend revenue (£423,883) is lower than Tuesday alone (£232,427), and order counts are unusually flat across the week (~1,370–1,499 per day). Saturday has the highest AOV but Sunday is the weakest day, pulling weekend performance down. This breaks the typical retail assumption that weekends dominate sales.

Business Recommendation

Stop scheduling marketing spikes purely around weekends — the data shows mid-week (Tuesday in particular) yields better marginal returns. Reallocate promotional and email-send cadence toward Tuesday–Thursday. Investigate the Saturday-vs-Sunday AOV gap to understand whether Sunday weakness is a checkout-experience issue or a demand-quality issue, and run a controlled test of Sunday-specific incentives.

14.12 Business Question 12

Question: Are customer acquisition costs rising faster than customer lifetime value, reducing growth profitability?

Analysis Performed

Computed blended CAC per year from marketing_spend (avg CAC, total spend, customers acquired), cohort LTV per customer by signup year from customers joined to Completed orders, and joined the two to derive the LTV-to-CAC ratio by cohort year. Also broke CAC out by channel and year.

Key Findings

Year	Avg LTV / Customer	Blended CAC	LTV-to-CAC Ratio
2022	£331.26	£24.70	13.41
2023	£315.64	£34.86	9.05
2024	£199.65	£49.21	4.06

Business Insight

Unit economics are collapsing: LTV:CAC dropped from 13.4 to 4.1 in two years — a dual-pressure failure where CAC nearly doubles while LTV falls by 40%. The 2024 cohort is approaching the warning zone (<5×). CAC inflation is universal across every channel, and Paid Social (the largest budget line) shows the steepest rise (£48.02 → £95.69). Scaling spend in 2024 is in a negative marginal-ROI regime.

Business Recommendation

Cap marketing spend growth until LTV:CAC recovers: the 2024 ratio of 4.06 means each incremental £1 of acquisition cost is producing only ~£4 of lifetime value, before accounting for returns and discounts. Pair the cap with the Q3 budget reallocation, the Q1 retention programme (to lift LTV) and the Q10 supplier fix (to reduce refund-driven LTV erosion). Set a blended LTV:CAC floor of 6× as a spend-release trigger; do not scale Paid Social again until its channel-level LTV:CAC exceeds 8×.

15. Recommendations

The recommendations below are organised by theme and directly address the twelve business problems analysed in Section 14. Each recommendation references the supporting analysis.

15.1 Marketing

- Reallocate spend from Paid Social and Google Ads (84.3% of budget, LTV:CAC 4.31 and 4.65) toward Email (33.41) and Organic (24.52). Move 10–15 percentage points of budget share in a phased shift. [Addresses Q3, Q12]
- Set a blended LTV:CAC floor of 6× and a per-channel floor of 8×; do not scale any channel below its floor. Cap total marketing spend growth until the ratio recovers. [Addresses Q12]
- Stand up a structured post-purchase email lifecycle programme to convert one-time buyers (68.8% of the base) into repeat buyers. [Addresses Q1]

15.2 Product & Supplier Quality

- Issue a 90-day supplier-quality improvement plan to NovaTech Supply with quantified defect-rate targets; suspend or reduce orders of Bluetooth Speaker Mini and Wireless Earbuds Pro pending improvement. [Addresses Q10, Q8]
- Review unit economics of the two loss-making products (Noise Cancel Headphones, Power Bank 20000mAh): renegotiate cost, raise price or discontinue. [Addresses Q2]
- Tie supplier scorecards to return-rate ceilings (target ≤19% overall within 12 months). [Addresses Q8]

15.3 Inventory

- Raise peak-season safety stock for the top five revenue-impact products (27in Monitor 4K, Smart Watch Fitness, Mechanical Keyboard TKL, Docking Station USB-C, Bluetooth Speaker Mini); target a 50% reduction in peak stockout days for these SKUs. [Addresses Q4]
- Incorporate revenue impact (price × daily velocity) into replenishment planning, not just unit-velocity forecasts. [Addresses Q4]
- Continue monitoring the Fast-Mover categories (Accessories, Peripherals, Audio) which dominate turnover and stockout exposure. [Addresses Q18]

15.4 Pricing & Discounting

- Replace blanket ~20% month-end discounting with behaviour-triggered lifecycle promotions; set a monthly discount-cost cap. [Addresses Q6]
- Track discount revenue-efficiency (discount £ per incremental revenue £) and target a reduction in month-end discount cost from £73,663 toward the rest-of-month baseline. [Addresses Q6]

15.5 Shipping & Order Economics

- Introduce a minimum free-shipping threshold (~£30) and tiered shipping fees for low-basket orders, especially in Cardiff and Scotland. [Addresses Q5, Q9]
- Add "frequently bought together" bundling modules (supported by the Q16 product-affinity analysis) to lift single-item baskets above the unprofitable shipping zone. [Addresses Q9]

- Track the count of orders where shipping $\geq$ subtotal as a weekly operational KPI; reduce the current 42 to zero. [Addresses Q9]

15.6 Customer Retention & Segmentation

- Launch a VIP programme for the top 500 (decile-1) customers with dedicated benefits; monitor days-since-last-order and trigger reactivation for outliers beyond ~250 days. [Addresses Q7]
- Build a mid-tier migration programme to convert high-frequency Non-VIPs into VIPs — the biggest upside is migration, not just protection. [Addresses Q7]
- Target the 500 At-Risk RFM customers (LTV £481.17, avg recency 709 days) with personalised reactivation. [Addresses Q13]

15.7 Demand Timing

- Reallocate promotional and email-send cadence toward Tuesday–Thursday, where demand is strongest; stop defaulting to weekend-centric campaigns. [Addresses Q11]
- Investigate the Saturday-to-Sunday AOV drop with a controlled test of Sunday-specific incentives. [Addresses Q11]

15.8 Recommendations Summary

#	Business Problem	Primary Recommendation	Expected Lever
1	Customer churn	Post-purchase retention programme	Lift 2nd-order rate
2	Loss-making products	Renegotiate/reprice or discontinue	Restore positive net profit
3	Marketing misallocation	Shift budget to Email/Organic	Higher blended LTV:CAC
4	Peak-season stockouts	Raise safety stock for top SKUs	Recover >£1.5M lost revenue
5	Regional shipping costs	Region-specific shipping thresholds	Protect Wales/Scotland margin
6	Month-end discount addiction	Lifecycle promotions + monthly cap	Reduce £73K discount leakage
7	VIP customer attention	VIP programme + mid-tier migration	Protect £640K VIP revenue
8	Rising return rate	Supplier scorecards + return ceiling	Return rate $\leq$ 19% within 12 months
9	Unprofitable small orders	Free-shipping threshold + bundling	Eliminate 42 ship $\geq$ subtotal orders
10	NovaTech quality issues	90-day quality plan + SKU suspension	Cut £101K NovaTech refunds
11	Weekend demand gap	Mid-week campaign reallocation	Better marginal ROI on promotions
12	Collapsing unit economics	Spend cap + 6 $\times$ LTV:CAC floor	Stop negative marginal-ROI spend

16. Conclusion

RetailIQ demonstrates a complete end-to-end analytics workflow — from cleaning a deliberately dirty eight-table dataset through relational modelling, SQL-based diagnosis of twelve business problems, and a seven-page Power BI executive dashboard. Every figure in this report is sourced directly from the project files: the cleaned CSV datasets, the eight Python EDA notebooks, the six MySQL scripts (with their embedded query outputs) and the published Power BI dashboard.

The analysis reveals a business with healthy top-line revenue but deteriorating unit economics. Three structural pressures — rising acquisition costs, falling per-customer value, and growing return and refund exposure — are compressing margin simultaneously. The good news is that the largest levers are identifiable and fixable: reallocate marketing spend, enforce supplier quality with NovaTech Supply, raise safety stock for a small set of premium high-velocity products, replace structural month-end discounting with lifecycle promotions, and introduce minimum-order-value shipping thresholds.

Implemented together, these interventions address the root causes identified across all twelve business problems rather than treating symptoms. The recommendations are prioritised by financial impact and are each traceable to a specific SQL finding or dashboard visual, making them suitable for a measurable, time-bound improvement plan.

— End of Report —